

UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy
DEGREE OF M.Sci. WITH HONOURS
FOURTH-YEAR FINAL EXAMINATION

03 30682

LM RELATIVISTIC ASTROPHYSICS

SUMMER EXAMINATION 2024

Time Allowed: 2 hours

Answer four questions from Section 1 and two questions from Section 2.

Section 1 consists of four questions and carries 40% of the marks for the examination.
Answer ***all four*** questions from this Section.

Section 2 consists of three questions and carries 60% of the marks.
Answer ***two*** questions from this Section. If you answer more than two questions,
credit will only be given for the best two answers.

The approximate allocation of marks to each part
of a question is shown in brackets [].

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text.
Calculators with the ability to store text should have their memories deleted prior to
the start of the examination.

A formula sheet and two tables of physical constants and units that may be required
will be found at the end of this question paper.

SECTION 1

Answer **all four** questions from this Section.

1. A star of radius R is entirely composed of a gas of N degenerate neutrons at zero-temperature. Each neutron has a total energy (the sum of the Fermi energy and the gravitational potential energy)

$$E \sim \hbar c \frac{N^{1/3}}{R} - G m_n^2 \frac{N}{R},$$

where m_n is the neutron's mass.

Estimate the value of the maximum mass and radius that this star can have while remaining in equilibrium.

[Hint: The condition $E = 0$ determines the maximum number of neutrons that constitute the star in equilibrium.]

[10]

2. A compact object of mass m orbits a black hole of mass $M_{\text{BH}} \gg m$ with orbital frequency f given by

$$f = \left(\frac{1}{2\pi} \right) \frac{M^{1/2}}{a^{3/2}}, \quad [c = G = 1],$$

where a is the orbit's semi-major axis and $M = M_{\text{BH}} + m$ the binary's total mass.

The orbit has a small eccentricity, $e \ll 1$, and the periastron precesses at a rate

$$\frac{d\omega_p}{dt} = 3 \frac{(2\pi f)^{5/3} M^{2/3}}{(1 - e^2)}, \quad [c = G = 1].$$

Calculate the semi-major axis of the orbit, in units of M , such that the precession frequency exceeds the orbital frequency, i.e. the periastron sweeps an angle of 2π in a time shorter than the orbital period. Discuss whether an actual astrophysical binary could be in such an orbit.

[10]

ANY CALCULATOR

3. Theories that suggest that primordial black holes may account for some fraction η of the dark matter mass-energy content of our Universe are back *en vogue*.

Under this assumption, and further assuming that primordial black holes have all the same mass, M_{PBH} , derive an order of magnitude estimate of the distance of the closest primordial black hole to the Earth as a function of η and M_{PBH} . Clearly state the assumptions you make.

You may assume that the critical energy density at the present epoch is:

$$\rho_{c,0} = 1.4 \times 10^{11} M_{\odot} \text{Mpc}^{-3}. \quad [10]$$

4. A white dwarf with a mass of $0.4 M_{\odot}$ has a radius of 10^3 km.

Calculate the fractional frequency shift due to gravitational redshift for an emission line produced by processes on the surface of this star.

The star is captured by a $10^9 M_{\odot}$ black hole on a highly eccentric orbit. Will the star survive intact until it plunges into the black hole or will it be tidally disrupted before it crosses the black hole horizon? Justify your answer quantitatively. [10]

SECTION 2

Answer **two** questions from this Section. If you answer more than two questions, credit will only be given for the best two answers.

5. A team of astronomers has just reported the discovery of a binary system of compact objects of component masses $m_1 = 2.5 M_\odot$ and $m_2 = 1.4 M_\odot$ with an orbital period $P = 6.3 \times 10^5$ s.

P is related to the binary semi-major axis, a , by $(2\pi/P)^2 = M/a^3$ in units in which $c = G = 1$. The binary's total mass, M , reduced mass, μ , and chirp mass, $\mathcal{M}$, are related to m_1 and m_2 according to the following expressions:

$$M = m_1 + m_2, \quad \mu = m_1 m_2 / M, \quad \mathcal{M}^{5/3} = \mu M^{2/3}.$$

Assume that the binary's orbit evolves purely due to gravitational wave emission and that the orbit is circular.

The rate at which energy is lost due to gravitational-wave emission is:

$$\frac{dE}{dt} = -\frac{32}{5}(\pi f \mathcal{M})^{10/3}, \quad [c = G = 1],$$

where f is the gravitational-wave frequency.

The rate of change of the gravitational-wave frequency is:

$$\frac{df}{dt} = \frac{96}{5}\pi^{8/3}\mathcal{M}^{5/3}f^{11/3}, \quad [c = G = 1].$$

- (a) Calculate the orbital separation of the two compact objects in AU . [3]
- (b) Calculate the gravitational-wave frequency at which the binary is radiating. [2]
- (c) Evaluate the remaining lifetime of the binary, i.e. the time the binary takes to coalesce. Express the result in years. [10]
- (d) Evaluate how many solar masses of equivalent energy are radiated in gravitational waves from now until the time the binary has an orbital period that is a half of its period today. [10]
- (e) Estimate the maximum value of the gravitational wave frequency radiated by this binary. Clearly state your assumptions. [5]

6. A rotating neutron star of mass M with magnetic field B and mean equatorial radius R loses energy through magnetic dipole radiation according to

$$\frac{dE}{dt} = -\frac{B^2 R^6 (2\pi f)^4 \sin^2 \alpha}{6c^3},$$

and through gravitational radiation according to

$$\frac{dE}{dt} = -\frac{32 G}{5 c^5} I^2 \epsilon^2 (2\pi f)^6.$$

Here f is the star's rotation frequency, α the angle between the magnetic field and the rotation axis, I the moment of inertia and ϵ the ellipticity. The rotational energy of the star is $E = I (2\pi f)^2 / 2$

- (a) Derive the expression for the characteristic age (the spin-down age) of this neutron star, $\tau \equiv -(f/\dot{f})$, in the case of energy loss dominated by:

i. magnetic braking; [4]

ii. gravitational wave emission. [3]

- (b) The neutron star is detected as a galactic radio pulsar. The measured values of the spin frequency and frequency derivative are $f = 44 \text{ Hz}$ and $\dot{f} = -3.4 \times 10^{-15} \text{ Hz s}^{-1}$.

Calculate the spin-down age of this neutron star assuming magnetic braking is the dominant physical process driving the energy loss. [3]

- (c) Gravitational waves produced by this rotating neutron star arrive at the Earth with amplitude

$$h = \frac{4\pi^2 G}{c^4} \frac{(I\epsilon)}{d} (2\pi f)^2,$$

where d is the distance to the neutron star.

- i. Derive an expression for the maximum value of h , as a function of τ , I and d . [8]

ii. Explain why this is an upper-limit to the actual gravitational-wave amplitude.

[2]

ANY CALCULATOR

- iii. Hence estimate the maximum value of the gravitational wave amplitude of this neutron star using typical values for a neutron star's mass and radius and the distance of radio pulsars. **[5]**
- (d) The rotation frequency is observed to suddenly increase by $3 \times 10^{-7} \text{ Hz}$ due to a glitch. Estimate the fractional change of the radius of the neutron star assuming no mass is lost during this episode. **[5]**

ANY CALCULATOR

7. A heavy neutron star of mass $M = 2.4 M_{\odot}$ and radius $R = 13 \text{ km}$ accretes material at a rate $dM/dt = 10^{-12} M_{\odot} \text{ yr}^{-1}$. Through this process energy is converted into radiation with efficiency $\eta = 0.1$.

- (a) Calculate the resulting accretion luminosity L_{acc} , and express the result in units of the solar luminosity $L_{\odot}$. [6]
- (b) Estimate the equivalent black-body radiation temperature associated with the energy produced through this accretion process. [4]
- (c) If accretion takes place in a plasma with sufficient density, the outward radiation pressure will balance the gravitational force onto the accreting plasma. Derive an expression for the associated luminosity (the Eddington limit) and compute its value for this heavy neutron star. [The Thomson cross-section has a value of $\sigma_T = 6.65 \times 10^{-29} \text{ m}^2$.] [11]
- (d) Estimate the time needed to radiate away an amount of energy equivalent to the rest mass of the neutron star if accretion takes place at 10% of the Eddington limit throughout the entire period. [6]
- (e) Estimate now the time needed to radiate away an amount of energy equivalent to a $10^8 M_{\odot}$ black hole if accretion takes place at constant rate of 10% of the Eddington limit onto a $10^8 M_{\odot}$ black hole. [3]

Formula Sheet

Relevant formulae are provided as part of the questions.

Astrophysical Constants and Units

Astronomical Unit	AU	$1.50 \times 10^{11} \text{ m}$
Parsec	pc	$3.1 \times 10^{16} \text{ m}$
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_{\odot}$	$3.84 \times 10^{26} \text{ W}$
Absolute bolometric magnitude of the Sun	M_{bol}	$+4^{\text{m}}75$
Bolometric correction for the Sun	BC	$-0^{\text{m}}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\text{m}}74$
Solar constant	f	$1.36 \times 10^3 \text{ W m}^{-2}$
Solar mass	$M_{\odot}$	$1.989 \times 10^{30} \text{ kg}$
Wien constant	b	$2.898 \times 10^{-3} \text{ m K}$
Hubble constant	H_o	$70 \text{ km s}^{-1} \text{ Mpc}^{-1}$
Solar radius	$R_{\odot}$	$6.95 \times 10^8 \text{ m}$
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_{\oplus}$	$5.972 \times 10^{24} \text{ kg}$
Earth radius	$R_{\oplus}$	6371 km

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
[<i>N.B.</i> 1 mole $\equiv$ 1 <i>gram-molecule</i>]		
Gas constant	R	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$
Stefan constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Rydberg constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$
	$h/2\pi$	$\hbar$ $1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$
Speed of light <i>in vacuo</i>	c	$2.998 \times 10^8 \text{ m s}^{-1}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	$1.602 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Rest energy of electron		0.511 MeV
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Rest energy of proton		938.3 MeV
One atomic mass unit	u	$1.66 \times 10^{-27} \text{ kg}$
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Magnetic constant	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12} \text{ m}$
Bohr radius	a_0	$5.2918 \times 10^{-11} \text{ m}$
angstrom	$\text{\AA}$	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	$133.32 \text{ Pa (N m}^{-2}\text{)}$

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.